Homework 6: Solutions

Leverage

1 Financial Literacy

Answer these questions in 3 steps: (1) give an answer to the question, (2) look over the material assigned for class and find a definition, (3) modify, if need be, your first answer. The goal is not for you to memorize a given answer but to be sure you can explain the concept to someone. If you can't, then you do not control that concept.

a) Risk Factor: A systematic variable, such as market beta, company size, or momentum, that helps explain differences in stock returns and investment risk across assets.

b) Equity as Residual Value: Value for shareholders after all liabilities have been paid.

c) Liquidation Value: Amount a company's assets would be worth if they were sold quickly, e.g. in the event of bankruptcy or closure.

d) Value at Risk (VAR): Shows how much money you could lose over a set time horizon at a given confidence level, usually 95%. It helps you assess the risk of not reaching your wealth goal. For example, if your VaR is $5,000 at 95% confidence, there is a 5% chance you could lose more than $5,000 over that period.

e) Prudential Regulation: Rules designed to ensure that financial institutions remain financially sound, limiting excessive risk-taking and reducing the probability of failure.

2 Thought Problems

Part 1

You have just woken up in a binomial world and you are a banker. You face many firms with desired capital K. They can secure this either through equity or from a loan from you.

Their gross asset value V evolves as a binomial. That means that next year their gross asset value will be either $V_u = (1 + r_f)V + v$ or $V_d = (1 + r_f)V - v$.

1. What is the present value of the firm if the market interest rate is r_b?

   The expected value at $t = 1$ is:

$$E[V_1] = \frac{V_u + V_d}{2} = \frac{(1 + r_f)V + v + (1 + r_f)V - v}{2} = (1 + r_f)V$$

   With r_b as market rate, the present value is:

$$PV_0 = \frac{(1 + r_f)V}{1 + r_b}$$

2. Firms come to ask you for a loan of value L. How much can you lend them if you want to be paid $(1 + r_b)L$ in every state of the world? How does that change as a function of v and as a function of $r_f - r_b$?

   Riskless repayment means that firms can pay in both state of the world. We have:

$$(1 + r_b)L = \min(V_u, V_d) = V_d = (1 + r_f)V - v$$
$$L_{\max} = \frac{(1 + r_f)V - v}{(1 + r_b)}$$

   - $L_{\max}$ is decreasing in v:

$$\frac{\partial L_{\max}}{\partial v} = -\frac{1}{1 + r_b} < 0.$$

   - As $r_f - r_b$ increases, $L_{\max}$ increases because the numerator rises with r_f and the denominator falls with r_b. Intuitively: higher asset growth and/or cheaper discounting supports more safe lending.

3. Now consider the same question after one period. What is the largest loan available if the binomial outcome is $V^1 = (1 + r_f)V - v$, or $V^1 = (1 + r_f)V + v$. What is the general insight about access to capital and returns?

At $t = 1$, the same logic applies with V_1 as the new "today" value. Value at $t = 2$:

$$V_2 = \begin{cases} (1 + r_f)V_1 + v \\ (1 + r_f)V_1 - v \end{cases}$$

So the maximum riskless loan at time 1 is:

$$L_{1,\max}(V_1) = \frac{(1 + r_f)V_1 - v}{1 + r_b}.$$

If the bad state occurred $V_1 = V_d = (1 + r_f)V - v$. Then:

$$L_{1,\max}(V_d) = \frac{(1 + r_f)\big((1 + r_f)V - v\big) - v}{1 + r_b}$$

If the good state occurred $V_1 = V_u = (1 + r_f)V + v$. Then:

$$L_{1,\max}(V_u) = \frac{(1 + r_f)\big((1 + r_f)V + v\big) - v}{1 + r_b}$$

Of course, $L_{1,\max}(V_u) > L_{1,\max}(V_d)$. It shows that access to safe capital is procyclical. A higher V_1 increases collateral in the worst next-period state, raising borrowing capacity.

4. Now suppose the bank is willing to take on some risk. What is the largest loan it can make and have an expected return of r_b ? What happens to equity when the binomial outcome is $V_d = (1 + r_f)V - v$.

   Now, we can use the expected value for V_1 instead of the lower bound:

$$(1 + r_b)L = E[V_1] = (1 + r_f)V$$
$$L_{\max} = \frac{(1 + r_f)V}{(1 + r_b)}$$

   The maximum loan is larger by $\frac{v}{1+r_b}$ compared to the riskless case.

   Assuming a face value $F = L_{\max}(1 + r_b)$, equity at $t = 1$ is $E_1 = \max\{V_1 - F, 0\}$. If the bad state occurs, this is negative so $E_1(V_d) = 0$, i.e. equity wiped out.

Part 2

You are an investor who is offered a pure long strategy (investment mix). Its period return is drawn from a uniform distribution between $\underline{r}$ and $\overline{r}$. The financial intermediary is willing to lend to you at r.

1. What are the expected returns of the pure long, the 1×, 2×, and 3× leveraged positions as a function of r. With unlimited liability what is the range of interest rates where you might want to be leveraged and what range of interest rates where a risk neutral person would not want to take on leverage.

   The strategy return is $r_s \sim U[\underline{r}, \overline{r}]$. Hence $E[r_s] = \frac{\underline{r}+\overline{r}}{2} \equiv \mu$. Assume money invested is 1. If you lever $L\times$, you borrow $L-1$ at rate r and invest L. Your net return on equity is: $R_L = Lr_s - (L-1)r$. Taking expectations: $E[R_L] = L\mu - (L-1)r$.

   The expected return of leverages positions are:

   - 1× (no leverage): $E[R_{1\times}] = \mu$
   - 2×: $E[R_{2\times}] = 2\mu - r$
   - 3×: $E[R_{3\times}] = 3\mu - 2r$

   Comparing $L\times$ to 1× we have $E[R_L] - \mu = (L-1)(\mu - r)$. So:

   - If $\mu > r$, leverage increases expected return
   - If $\mu < r$, leverage reduces expected return

   With unlimited liability, a risk-neutral investor:

   - If $\mu > r$, wants infinite leverage
   - If $\mu < r$, wants no leverage
   - If $\mu = r$, is indifferent

2. No suppose there is limited liability (or that you have no other wealth). Does that change your answer?

   Yes, it does. With limited liability, you cannot loose more than what you invested, in this case 1. Your return is:

$$R_L^{LL} = \max\{Lr_s - (L-1)r,\ -1\}$$

When there is a positive probability that $Lr_s - (L-1)r < -1$, which depends on $\underline{r}$, an investor with limited liability may be willing to leverage even if $\mu < r$ because his losses are capped. If $\underline{r}$ is such that there is no probability that $Lr_s - (L-1)r < -1$, then it doesn't matter.

3. Is taking on 2× leverage on half your wealth and going pure long on the other half the same as taking on 1× leverage. Explain your answer. What lesson about risk do you draw from this?

   The mixed strategy return is:

$$R_{\text{mixed}} = \frac{1}{2}(2r_s - r) + \frac{1}{2}r_s = \frac{3}{2}r_s - \frac{r}{2} = R_{1.5\times} \neq R_{1\times}$$

   Leverage scales linearly with exposure and cannot be offset by mixing leveraged and unleveraged positions.

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Part 3

In recent years, some investors have bought a lot of housing in a small number of US cities where the rent to price ratio is high, the funds are highly leveraged with 3-5 year loans. An investor comes to you and asks you to advise on the suitability of this strategy as a dominant part of her portfolio. As a financial advisor you must educate your client about potential returns and potential risk. Given that this is a leveraged position you need to what influences the equity value.

1. Please write an equation that would allow your client to understand the NPV of this fund if all the houses are sold after 5 years.

Let:

- P_0 = total purchase price of houses
- D_0 = debt
- $E_0 = P_0 - D_0$ = initial equity
- r = borrowing rate
- R_t = rental income in year t
- g = house price appreciation rate
- $P_5 = P_0(1+g)^5$ = sale value in 5 years
- k_E = required return on equity

Equity cash flows:

- Each year, rental income minus debt payment: $CF_t^{equity} = R_t - rD_0$
- At year 5: sale value minus remaining debt $CF_5^{equity} = P_5 - D_0$

So the equity NPV is:

$$NPV = \sum_{t=1}^{5} \frac{R_t - rD_0}{(1+k_E)^t} + \frac{P_5 - D_0}{(1+k_E)^5} - E_0$$

2. Given that equation, you can describe the upside. What set of key changes in the US economy would lead to higher than expected returns

   Some examples are:

   - Stronger national income/employment increasing demand for housing: $R_t \uparrow$
   - Fed cuts rates implying lower financing costs (if floating rates on debt): $r \downarrow$
   - Easier credit conditions on mortgages increasing demand and housing prices: $P_5 \uparrow$
   - Lower risk premia on equity implying lower discounting: $k_E \downarrow$

3. Given that equation, you can describe the upside. What set of key changes in the economy of the cities where the housing is located would lead to higher than expected returns.

   Some examples are:

   - Strong local job growth or in-migration increasing housing demand: $R_t \uparrow$
   - Local supply constraints (e.g. land scarcity) raising rents and prices: $R_t \uparrow$, $P_5 \uparrow$
   - Infrastructure improvements increasing property values: $P_5 \uparrow$
   - Lower perceived local risk (e.g. neighborhood is safer): $k_E \downarrow$

4. Given that equation, you can describe the downside. What set of key changes in the US economy would lead to lower than expected returns

   Take the opposite of changes discussed in Part 2.

5. Your client then decides that she wants to find a fund has the same strategy but is equity only. What is the tradeoff she is willing to make.

   An equity-only version sets $D_0 = 0$. In the NPV expression, which lowers downside risk and eliminates the possibility that equity is wiped out when P_5 falls close to D_0.

   The tradeoff is that she gives up the leverage amplification of returns: with $D_0 = 0$ the equity payoff is not boosted by borrowing, so expected returns are lower but the investment is safer (lower volatility and no default risk).

3 CAPM and Your Portfolios

Take all the stocks you have picked so far.

The answer to this question will depend on the stocks you picked. We provide here a mock example with Tesla, Nvidia and Walmart.

1. Look up their market betas on a finance website. What is the distribution of beta that you have picked? What does it say about your expectations of aggregate market performance?

   We use here data from: https://finance.yahoo.com/. Yahoo reports the Beta 5Y Monthly, which is calculated using data from the last 5 years at the monthly level.

   The beta for our stocks are:

   - Tesla: 1.89
   - Nvidia: 2.31
   - Walmart: 0.67

   A beta > 1 means that the stock is more sensitive to market movements than the overall market. Tesla and Nvidia are substantially more sensitive to market fluctuations, which is typical for tech companies. Walmart has a beta below 1, meaning it is less sensitive to market movements, which makes sense because it sells basic goods that consumers continue to buy even in weaker economic conditions. The distribution of betas in our portfolio is skewed toward high-beta stocks. Overall, we would expect the portfolio to be more sensitive to aggregate market performance than the market itself. It would likely outperform the market in strong market periods but underperform in downturns.

2. For three of the stocks you picked, download their prices for the last year and the S&P500 index. Compute their market betas by normalizing all prices at the beginning of the sample to 100. Estimate the covariance of your stocks with the S&P 500.

   You'll find below a Python code achieving this for the three stocks we picked. You can use the yahoo package directly in Python to avoid downloading the data. The results:

Table 1: Estimated Daily Market Betas

Stock	Covariance	Beta
TSLA	0.0002	2.3653
NVDA	0.0002	2.1864
WMT	0.0000	0.4784

```python
import yfinance as yf
import pandas as pd

data = yf.download(["TSLA", "NVDA", "WMT", "^GSPC"],
                   start="2024-02-16",
                   interval="1d",
                   auto_adjust=True)

# Auto-adjust accounts for dividends, etc.

prices = data["Close"]

normalized = prices / prices.iloc[0] * 100

returns = normalized.pct_change().dropna()

market_var = returns["^GSPC"].var()

covariances = {}
betas = {}

for stock in ["TSLA", "NVDA", "WMT"]:
    cov = returns[stock].cov(returns["^GSPC"])
    beta = cov / market_var
    covariances[stock] = cov
    betas[stock] = beta

results = pd.DataFrame({
    "Covariance": covariances,
    "Beta": betas
}).round(4)
```

```
print(results)
```

3. How close do you get to the finance website? (You may not get the same number but they should pretty much be in the same rank order).

   The differences from Part 1 arise from the fact that we use daily data over the past year, whereas Yahoo reports a 5-year beta computed using monthly returns. Tesla, for instance, experienced large price movements over the last year, which have a strong impact on the estimated beta in a short daily sample, but are averaged out in a longer 5-year monthly sample. Overall, the replication works reasonably well.

4 The Market for Lemons

The Happy Valley bank is making business loans. When a business fails, it can make claims on the business but not on the wealth of the owners. Initially, there is one type of business. It has a capital stock that grows after two years to 600,000 if the firm succeeds and to 150,000 if it fails (the probability of success is 90%). To start, the owners ask for a loan of X. The bank needs to make an expected return of 8%. Assume the businesses' success probabilities are independent.

1. Create a formula that tells the bank what interest to charge given X so that its portfolio of loans has a return of 8%.

   - If $X(1.08) > 600,000$, the required repayment exceeds the firm's maximum value, so the bank does not offer the loan.
   - If $150,000 < X(1.08) < 600,000$, the bank's expected return equation is:

     $$0.1(150,000) + 0.9(X)(1+r) = X(1.08)$$

     Solving for r:

     $$r = \frac{X(1.08) - 15,000}{0.9X} - 1$$

   - If $X(1.08) < 150,000$, the loan is covered in either state, and the bank offers the loan at 8%.

2. Assume the firms all borrow \$200,000. Now a second type of firm enters. Its owners also borrow \$200,000 and if they succeed, they grow to \$500,000. If they fail, there is no residual value (they fail half the time). If the bank can tell the two types of firms apart, what interest rate(s) does it charge this second group?

- Type 1 Firm (we can use the formula from the previous part):

$$r_1 = \frac{200,000(1.08) - 15,000}{0.9(200,000)} - 1 = 0.1166$$

- Type 2 Firm (50% success): We must calculate the formula for r_2 from

$$0.5(200,000)(1 + r_2) = X(1.08)$$

implying

$$r_2 = \frac{200,000(1.08)}{0.5(200,000)} - 1$$

$$r_2 = \frac{216,000}{100,000} - 1 = 1.16$$

3. Assume now the bank can't tell the two types of firms apart. What is the market equilibrium if there are 10% high-risk firms?

- Probability of receiving \$150,000 (low risk firm failing): $0.9 \times 0.1 = 0.09$
- Probability of receiving more than \$150,000 (low and high risk firms succeeding): $(0.1 \times 0.5) + (0.9 \times 0.9) = 0.86$
- Solving for r: $0.09 \cdot 150,000 + 0.86 \cdot 200,000(1 + r) = 200,000(1.08)$

$$r = \frac{216,000 - 13,500}{172,000} - 1 = 0.1773$$

4. What is the market equilibrium if there are 50% high-risk firms?

- Probability of receiving \$150,000 (low risk firm failing): $0.5 \times 0.1 = 0.05$
- Probability of receiving more than \$150,000 (low and high risk firms succeeding): $(0.5 \times 0.5) + (0.5 \times 0.9) = 0.7$
- Solving for r: $0.05 \cdot 150,000 + 0.7 \cdot 200,000(1 + r) = 200,000(1.08)$

$$r = \frac{216,000 - 7,500}{140,000} - 1 = 0.4892$$

5. Assume a diviner comes to town and charges \$10,000 per firm he examines to certify that it is a good firm or a bad one. If he is 90% accurate, will the bank want to hire him? Will a good firm hire the diviner? What interest rate will the bank charge to

certified firms?

Without certification from the diviner, a firm will be charged a high interest rate of 48.92%. A low-risk firm does not like it because it has to borrow at a higher cost because it is in the same pool as the high risk firm. We need to calculate the posterior.

- Probability of having a "good" label: $Pr(Good) = 0.5 \cdot 0.9 + 0.5 \cdot 0.1 = 0.5$, i.e. with 50% chance this is a low-risk firm and it has a good label with 90% chance, with 50% chance it's a high-risk firm and the diviner is wrong with 10% chance.
- Posterior: $Pr(Low\ Risk|Good) = \frac{0.5 \cdot 0.9}{0.5} = 0.9$; $Pr(High\ Risk|Good) = 0.1$

Therefore, a low-risk firm with certification will be charged Part 3 rate, i.e. 17.73%.

For a low-risk firm, the face value in the two cases are:

- $F_{\text{No Cert}} = 200,000(1 + 0.4892) = 297,840$
- $F_{\text{Cert}} = 200,000(1 + 0.1773) = 235,460$

If the low-risk firm fails, liquidation value goes into debt service and it doesn't earn anything. If it succeeds, it earns the difference in face value. Therefore, it expects to gain $0.9(297,840 - 235,460) = 56,142$ being certified, which is higher than $10,000$. It will want to be certified.

A risk neutral bank doesn't care about the certification because it charged the break-even, competitive rate in each case. So, unless it can increase rates to absorb the $\$10,000$ fee, it won't hire the diviner.

5 Value at Risk

UIT, the famous institute, receives a gift of \$600 million. It invests in a fund that returns 21% in good years and -3% in bad years. The institute withdraws 5% a year to pay for expenses. Assume good and bad years are independent.

1. What are the possible fund values after one, two, three, four, and five years? How likely is each outcome? What is the probability the fund is worth less than \$500 million after five years? If it is worth less than \$500 million, what is its expected value?

2. What is the value at risk that the CIO can report with 80% confidence, and what is the expected value of the fund if it is at or below the VaR?

3. The CIO contemplates three additional strategies: Increase the size of the fund by borrowing \$250 million, \$500 million, and \$1 billion at 2% interest. What is the expected annual return to the fund?

4. What are the possible fund values after one, two, three, four, and five years if the CIO borrows \$1 billion? How likely is each outcome? What is the probability the fund cannot pay back its loan after 1, 2, 3, 4, or 5 years?

5. Show that if the fund's first year is good, then it can always repay the loan.

6. Show that if the fund only borrows \$500 million, it can always repay the loan.

The solution is attached on the next page as a Jupyter notebook.

5) Value at Risk

UIT, the famous institute receives a gift of $600 million. It invests in a fund that returns 21% in good years and -3% in bad years. The institute withdraws 5% a year to pay for expenses. Assume good and bad years are independent.

In [1]:
```
%reset -f
%pip install pandas
```

```
Requirement already satisfied: pandas in /Library/
Frameworks/Python.framework/Versions/3.11/lib/
python3.11/site-packages (2.3.3)
Requirement already satisfied: numpy>=1.23.2 in /
Library/Frameworks/Python.framework/Versions/3.11/
lib/python3.11/site-packages (from pandas) (2.3.1)
Requirement already satisfied: python-
dateutil>=2.8.2 in /Users/guillaumewegmuller/
Library/Python/3.11/lib/python/site-packages (from
pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /
Library/Frameworks/Python.framework/Versions/3.11/
lib/python3.11/site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /
Library/Frameworks/Python.framework/Versions/3.11/
lib/python3.11/site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in /Users/
guillaumewegmuller/Library/Python/3.11/lib/python/
site-packages (from python-dateutil>=2.8.2->pandas)
(1.17.0)

[notice] A new release of pip available: 22.3.1 ->
26.0.1
[notice] To update, run: pip3 install --upgrade pip
Note: you may need to restart the kernel to use
updated packages.
```

In [2]:
```
import math
import pandas as pd
import itertools
```

a) What are the possible fund values after one, two, three, four, and five years. How likely is each outcome? What is the probability the fund is worth less than $500 million after 5 years? If it is worth less than $500 million what is its expected value?

In [3]:

```
# Parameters

V0 = 600 # $ millions
good_ret = 0.21
bad_ret = -0.03
p_good = 0.5
withdraw = 0.05 # 5% withdrawal each year

g_factor = (1 + good_ret) # 1.21
b_factor = (1 + bad_ret) # 0.97
w_factor = (1 - withdraw) # 0.95
```

In [4]:

```
# Likelihood of each outcome

def outcomes_at_t(t) -> pd.DataFrame:
    rows = []
    for g in range(t + 1):
        prob = math.comb(t, g) * (p_good ** g) *
((1 - p_good) ** (t - g))
        value = V0 * (w_factor ** t) * (g_factor
** g) * (b_factor ** (t - g))
        rows.append({"t": t, "good_years": g,
"value_millions": value, "prob": prob})
    df =
pd.DataFrame(rows).sort_values("value_millions").r
eset_index(drop=True)
    return df

for t in range(1, 6):
    print(outcomes_at_t(t))
```

```
   t  good_years  value_millions  prob
0  1           0           552.9   0.5
1  1           1           689.7   0.5
   t  good_years  value_millions  prob
0  2           0       509.49735  0.25
1  2           1       635.55855  0.50
2  2           2       792.81015  0.25
   t  good_years  value_millions   prob
0  3           0      469.501808  0.125
1  3           1      585.667204  0.375
2  3           2      730.574553  0.375
3  3           3      911.335267  0.125
   t  good_years  value_millions    prob
0  4           0      432.645916  0.0625
1  4           1      539.692328  0.2500
2  4           2      673.224451  0.3750
3  4           3      839.795449  0.2500
4  4           4     1047.579890  0.0625
   t  good_years  value_millions     prob
0  5           0      398.683212  0.03125
1  5           1      497.326481  0.15625
2  5           2      620.376331  0.31250
3  5           3      773.871506  0.31250
4  5           4      965.344869  0.15625
5  5           5     1204.193083  0.03125
```

In [5]:

```python
# Fund is worth less than $500 million after 5 years with probability

df5 = outcomes_at_t(5)
p_below_500 = df5.loc[df5["value_millions"] < 500, "prob"].sum()
print(f"Answer: {p_below_500:.4f}")
```

```
Answer: 0.1875
```

In [6]:

```python
# Expected value if worth less than $500 million after 5 years
# Using formula for conditional expectation: E[X | A] = (sum X_i P_i) / P(A)

mask = df5["value_millions"] < 500
ev_cond = (df5.loc[mask, "value_millions"] * df5.loc[mask, "prob"]).sum() / p_below_500 if p_below_500 > 0 else float("nan")
print(f"Answer: ${ev_cond:.2f} million")
```

```
Answer: $480.89 million
```

b) What is the value at risk that the CIO can report with 80% confidence and what is the expected value of the fund if it is at or below the VAR.

In [7]:
```
alpha = 0.80 # 80% confidence
tail = 1 - alpha # left-tail probability (20%)

df5_sorted =
df5.sort_values("value_millions").reset_index(drop
=True)
df5_sorted["cdf"] = df5_sorted["prob"].cumsum()

# VaR
var_80 = df5_sorted.loc[df5_sorted["cdf"] >= tail,
"value_millions"].iloc[0]
print(f"VaR: ${var_80:.2f} million")

# Expected Value if at or below the VaR
mask_var = df5_sorted["value_millions"] <= var_80
p_tail = df5_sorted.loc[mask_var, "prob"].sum()
ev_below_var = ((df5_sorted.loc[mask_var,
"value_millions"] * df5_sorted.loc[mask_var,
"prob"]).sum()/p_tail)
print(f"Expected Value: ${ev_below_var:.2f}
million")
```

```
VaR: $620.38 million
Expected Value: $568.07 million
```

c) The CIO contemplates three additional strategies. Increase the size of the fund by borrowing 250 million, 500 million, and 1 billion dollars at 2% interest. What is the expected annual return to the fund?

In [8]:
```
expected_growth = 1.09 # Expected return: 0.5*0.21
+ 0.5*(-0.03) = 0.09
loan_rate = 0.02

def expected_value_one_year(B):
    return ((V0 + B) * expected_growth - loan_rate
* B) * w_factor


# Borrowing scenarios
borrow_levels = [250, 500, 1000]

for B in borrow_levels:
    EV1 = expected_value_one_year(B)
    expected_return = EV1 / V0 - 1
    print(f"Borrowing ${B} million: Expected
return = {expected_return:.2%}")
```

```
Borrowing $250 million: Expected return = 45.90%
Borrowing $500 million: Expected return = 88.26%
Borrowing $1000 million: Expected return = 172.97%
```

d) What are the possible fund values after one, two, three, four, and five years if the CIO borrows 1 billion dollars? How likely is each outcome? What is the probability the fund cannot pay back its loan after 1,2,3,4 or 5 years?

In [9]:
```
B = 1000 # borrow $m
interest = 0.02 * B # 2% interest on the loan
V_start = V0 + B

def step(V, good):
    return (V * (g_factor if good else b_factor) -
interest) * w_factor

# Order of outcomes now matters because of
interest
def outcomes_at_t(t):
    rows = []
    for path in itertools.product([0, 1],
repeat=t): # 0=Bad, 1=Good
        V = V_start
        for s in path:
            V = step(V, s == 1)

        label = "".join("G" if s else "B" for s in
path)
        prob = (p_good ** sum(path)) * ((1 -
p_good) ** (t - sum(path)))
```

```
        rows.append({
            "t": t,
            "path": label,
            "value_millions": V,
            "prob": prob
        })

    return
pd.DataFrame(rows).sort_values("value_millions").r
eset_index(drop=True)


for t in range(1, 6):
    df = outcomes_at_t(t)
    print(df.to_string(index=False, formatters={
        "value_millions": "{:,.4f}".format,
        "prob": "{:.6f}".format
    }))

for t in range(1, 6):
    df = outcomes_at_t(t)
    p_cannot_repay = df.loc[df["value_millions"] <
B, "prob"].sum()
    print(f"t={t}: Probability it cannot repay:
{p_cannot_repay:.6f}")
```

```
 t path value_millions     prob
 1    B     1,455.4000 0.500000
 1    G     1,820.2000 0.500000
 t path value_millions     prob
 2   BB     1,322.1511 0.250000
 2   BG     1,653.9823 0.250000
 2   GB     1,658.3143 0.250000
 2   GG     2,073.3199 0.250000
 t path value_millions     prob
 3  BBB     1,199.3622 0.125000
 3  BBG     1,500.8127 0.125000
 3  BGB     1,505.1447 0.125000
 3  GBB     1,509.1366 0.125000
 3  BGG     1,882.2527 0.125000
 3  GBG     1,887.2323 0.125000
 3  GGB     1,891.5643 0.125000
 3  GGG     2,364.2812 0.125000
 t path value_millions     prob
 4 BBBB     1,086.2123 0.062500
 4 BBBG     1,359.6669 0.062500
 4 BBGB     1,363.9989 0.062500
```

```
 4 BGBB     1,367.9908 0.062500
 4 GBBB     1,371.6694 0.062500
 4 BBGG     1,706.1842 0.062500
 4 BGBG     1,711.1638 0.062500
 4 BGGB     1,715.4958 0.062500
 4 GBBG     1,715.7526 0.062500
 4 GBGB     1,720.0846 0.062500
 4 GGBB     1,724.0765 0.062500
 4 BGGG     2,144.6494 0.062500
 4 GBGG     2,150.3735 0.062500
 4 GGBG     2,155.3531 0.062500
 4 GGGB     2,159.6851 0.062500
 4 GGGG     2,698.7413 0.062500
 t  path value_millions     prob
 5 BBBBB       981.9446 0.031250
 5 BBBBG     1,229.6010 0.031250
 5 BBBGB     1,233.9330 0.031250
 5 BBGBB     1,237.9250 0.031250
 5 BGBBB     1,241.6036 0.031250
 5 GBBBB     1,244.9934 0.031250
 5 BBBGG     1,543.9371 0.031250
 5 BBGBG     1,548.9167 0.031250
 5 BBGGB     1,553.2487 0.031250
 5 BGBBG     1,553.5055 0.031250
 5 GBBBG     1,557.7340 0.031250
 5 BGBGB     1,557.8375 0.031250
 5 BGGBB     1,561.8294 0.031250
 5 GBBGB     1,562.0660 0.031250
 5 GBGBB     1,566.0579 0.031250
 5 GGBBB     1,569.7365 0.031250
 5 BBGGG     1,942.2587 0.031250
 5 BGBGG     1,947.9828 0.031250
 5 BGGBG     1,952.9624 0.031250
 5 GBBGG     1,953.2576 0.031250
 5 BGGGB     1,957.2944 0.031250
 5 GBGBG     1,958.2372 0.031250
 5 GBGGB     1,962.5692 0.031250
 5 GGBBG     1,962.8259 0.031250
 5 GGBGB     1,967.1579 0.031250
 5 GGGBB     1,971.1499 0.031250
 5 BGGGG     2,446.2745 0.031250
 5 GBGGG     2,452.8544 0.031250
 5 GGBGG     2,458.5784 0.031250
 5 GGGBG     2,463.5581 0.031250
 5 GGGGB     2,467.8901 0.031250
 5 GGGGG     3,083.2031 0.031250
t=1: Probability it cannot repay: 0.000000
t=2: Probability it cannot repay: 0.000000
```

```
t=3: Probability it cannot repay: 0.000000
t=4: Probability it cannot repay: 0.000000
t=5: Probability it cannot repay: 0.031250
```

e) Show that if the fund's first year is good, then it can always repay the loan?

In [10]:

```
# For years t ={1,2,3,4} it can always repay
# Need to show that if first draw is G, then it can always repay at t=5

df_Gfirst = df[df["path"].str.startswith("G")]

p_cond = (
    df_Gfirst.loc[df_Gfirst["value_millions"] < B, "prob"].sum()
    / df_Gfirst["prob"].sum()
)

print(f"Cannot repay at t=5 if first year good: {p_cond:.6f}")
```

```
Cannot repay at t=5 if first year good: 0.000000
```

f) Show that if the fund only borrows $500 million, it can always repay the loan?